

Contact

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Top Skills

Amazon Web Services (AWS)
Azure DevOps Services
Continuous Integration and
Continuous Delivery (CI/CD)

Languages

French (Native or Bilingual)
English (Native or Bilingual)
Danish (Elementary)

Publications

Characterization of postdive recovery using sound recordings and its relationship to dive duration, exertion and foraging effort of southern elephant seals (*Mirounga leonina*)

Elephant seals as ecosystem sentinels for the northeast Pacific Ocean twilight zone.

Digital evolution: Novo Nordisk's shift to ontology-based data management.

Variation in body condition during the post-moult foraging trip of southern elephant seals and its consequences on diving behaviour.

Adjustment of diving behaviour with prey encounters and body condition in a deep diving predator: the Southern Elephant Seal

Joffrey Joumaa

PhD | Data Analytics | Data Management | Digital Transformation |
Novo Nordisk
Denmark

Summary

I started my career tagging elephant seals in the Kerguelen Islands and applying machine learning to understand how they hunt, without ever seeing them do it. From field data to predictive models, that work set the tone for everything since.

Since then I've worked across agriculture, consumer goods, marine research, consultancy, and pharma as a data scientist and team lead, building models, dashboards, and data infrastructure that connect technical work to business decisions. I'm currently at Novo Nordisk, leading a cross-functional team working on data interoperability and knowledge infrastructure at scale.

I'm at my best when I'm close to both the technical problem and the business question, and I'm currently looking to move further into data analytics and business intelligence.

Experience

Novo Nordisk
2 years 8 months

Product Owner & Team Lead, Data Foundation
January 2025 - Present (1 year 3 months)
Copenhagen, Capital Region of Denmark, Denmark

- Leading a distributed, cross-functional team across 4 countries, carrying end-to-end responsibility for roadmap prioritisation, team delivery, and stakeholder alignment with 20+ partners. Following an internal reorganisation, my role operates in practice as much as a team lead as a product owner, including writing OKRs, defining job descriptions for new hires, and reporting directly to our Director.
- Driving the development of knowledge graphs built on ontology standards to improve how data flows and connects across systems such as ELN and

LIMS, with FAIR principles (Findable, Accessible, Interoperable, Reusable) as a guiding framework.

- Contributing to an ongoing, cross-functional effort to design and implement semantically enriched data contracts, improving how systems communicate and how data is governed across the organisation.
- Driving the team's delivery of controlled terminology aligned with existing ontologies, creating a shared data language that reduces inconsistency at the source.

Semantic Engineer

August 2023 - December 2024 (1 year 5 months)

Copenhagen, Région de la Capitale, Denmark

Built and maintained data infrastructure at the intersection of knowledge engineering and automation, working within Novo Nordisk's broader digital transformation.

- Designed and deployed pipelines in Azure DevOps to automate ontology testing on EC2-based scripts, reducing manual overhead and improving release reliability
- Built a proof-of-concept chat application using a graphRAG approach, enabling natural language querying over structured internal knowledge
- Stepped in as acting Product Owner during a transition period, keeping delivery on track across the team
- Contributed to the early framework for data contracts, helping shape how data is described, shared, and governed across systems
- Applied ontology embeddings to power semantic search, and began prototyping automated ontology reconciliation to reduce manual curation work

University of California, Santa Cruz

Data Scientist

October 2021 - July 2023 (1 year 10 months)

Santa Cruz, Californie, États-Unis

Turning raw information into valuable marine ecology research by leveraging data collected using new cutting edge tags (i.e. mainly motion sensors) deployed in northern elephant seals:

- Package development (R)
- Classification (DBSCAN, k-means)
- Dash application (dash, python)
- R package development
- Containerization (Docker)
- Supervision of undergraduate and PhD students

Thegreendata

Data Scientist

April 2020 - August 2021 (1 year 5 months)

Ville de Paris, Île-de-France, France

Developed econometric models and decision-support tools to help stakeholder in the agriculture sector becoming data-driven

- Classification (k-means, Random Forest, hierarchical clustering, CART)
- Dimensionality Reduction Algorithms (PCA)
- Prototyping (Shiny, Dash)
- Time series forecasting (ARIMA family)
- Regularized Regression (LASSO, Ridge)
- Simulation (Monte Carlo)

Saegus

1 year 5 months

Data Scientist - Groupama

January 2020 - March 2020 (3 months)

Paris et périphérie

- Leveraged data from call routing to improve and align customer experience among all regional entities (Spark, Dataiku)
- Transposed several conversion rate models initially developed with SAS, towards new solutions with Python and Dataiku.

Tableau Practice Lead - Bolloré Transport & Logistics

July 2019 - December 2019 (6 months)

Région de Paris, France

- Provided support and expertise on various projects initiated with Tableau (Desktop, Prep and Server)
- Helped with the creation of a Data Visualization pole within Bolloré
- Community manager to help with the adoption of Tableau

Data Analyst - L'Oréal

November 2018 - June 2019 (8 months)

Région de Paris, France

- Build reports and dashboards to monitor KPIs (Key Performance Indicators), understand drivers of KPI changes
- Data visualization creation, maintenance and support in a worldwide context (3500 + users, 64 countries)
- Implementation of an influencers monitoring page

IT&M STATS

Biostatistician - Da Volterra

July 2018 - October 2018 (4 months)

Région de Paris, France

- Implementation of statistical analysis for 16S metagenomic data (parametric and non-parametric tests, mixed models with random effects or repeated measures, adjustment for multiple testing)
- Management of subcontracting analyses

Malakoff Médéric

Data Analyst

December 2017 - June 2018 (7 months)

Région de Paris, France

Withing the Customer Service Department:

- Implementation of a customer call reminder segmentation based on text fields filled by phone operators
- Text Mining (Topic Modeling, Regular Expression, Cloudword)
- Data Visualization (ggplot2, Shiny, D3.js, Plotly)

CNRS - Centre national de la recherche scientifique

3 years 2 months

PhD

October 2013 - November 2016 (3 years 2 months)

Beauvoir sur Niort

By using machine learning to analyze bio-logging data collected from tagged southern elephant seals (*Mirounga leonina*), the objectives of my PhD were to investigate the diving and predation behaviours of the largest phocid in the world, without ever seeing it.

- Regression (Logistic, Linear, Additive, ...)
- Classification (Random Forest, K-means, KNN, SVM, ...)
- Statistics (Multivariate Analysis, PCA, ...)
- Signal Processing (Spectral Estimation, Signal Filtering)
- Optimal Foraging Theory, Optimal Diving Theory
- Movement Ecology (3D Trajectory Reconstruction using Dead Reckoning Methods, First Passage Analysis)

This PhD was carried out with financial support from the DGA (Direction Générale de l'Armement) and in collaboration with the SMRU (Sea Mammal Research Unit, University of St Andrews, Patrick Miller).

Fieldworker

September 2014 - November 2014 (3 months)

Kerguelen Islands

- Tagging and tag recovery on Southern elephant seals
- Elephant seals pups monitoring
- Field missions logistics (team organization, time management, ...)
- Capture, anesthesia and blood sample on elephant seals

Education

Université de La Rochelle

PhD, Quantitative Ecology · (2013 - 2016)

Université Paul Sabatier (Toulouse III)

Master 2 (M2), Biodiversité, Écologie, Évolution · (2012 - 2013)

Université de Bretagne Occidentale

Diplôme Universitaire : Découvertes du monde de l'entreprise, Biologie marine et océanographie biologique · (2011 - 2012)

Institut Univesitaire Européen de la Mer

Master 1 Sciences de la Mer et du Littoral, Sciences Biologiques
Marines · (2010 - 2011)

Université de La Rochelle

Licence Biologie - Biochimie, Biologie et Écologie Marines · (2009 - 2010)